Surprise-gated learning in a spiking network: one-page summary

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The question. When we hear a rare tone among frequent ones, neurons fire more. Is that just tired neurons (adaptation), or is the brain learning what to expect and signalling surprise (prediction error)? I also wanted to know what is the *smallest* piece of machinery that turns a network that only adapts into one that learns expectations quickly.

What I built. An open, fully reproducible spiking neural network (120 neurons, Brian2) in which every mechanism can be switched on and off on its own:

- two kinds of adaptation;
- a predictive-coding core with learned predictions and error neurons;
- my proposal, a **surprise gate**. After a prediction error it briefly speeds up learning, raises sensory gain and opens several candidate guesses about what comes next.

I wrote down every prediction and pass rule before running each experiment (pre-registration), ran 20 random seeds per condition, and published every result, including the predictions that failed. The repository has 79 unit tests, one command per figure, and cached, resumable runs.

What I found.

1. **Adaptation is not enough.** Adaptation alone reproduces the familiar "bigger response to the rare tone", but all of it is repetition suppression. The prediction-error component is zero, and there is no response at all to a *missing* tone. This holds even when adaptation is made 10 times slower.
2. **Surprise gating buys speed, not new abilities.** Two tests separate adaptation from expectation: an unexpected repeat (ABAB**AA**) and a missing tone. Adaptation gets the repeat backwards (−2.8 spikes). The surprise gate gets it right after normal training (+7.6 [4.0, 11.2]). The plain predictive core also gets there, but only with about 3 times more training. Ablations show that passing both tests needs prediction-error neurons, learning and the gate together.
3. **Honest limits.** One-shot learning, a predicted drop-off ordering and "accepting randomness" were *not* supported. When I slowed adaptation 3 times, the effects first disappeared; re-tuning by my original rules (a 15% change to three weights) brought them back. At 10 times slower, only a weak repeat effect returns.

Why it matters for AI. Standard training treats each example in much the same way, whether or not it was surprising. This model is a small worked example of a different rule: **learn fast exactly when the world surprises you, and keep a few competing hypotheses open while you do**. That is the core of sample-efficient, continual learning: adapt to a change in a handful of examples without overwriting everything else. The failures are informative too. The gate alone never stops reacting to pure noise, so a system also needs a sense of how predictable its input is (precision) before it can tell surprise from randomness. That is the next step I am building.

Paper (10 pages) and code: github.com/Aryan-sk6/SNNresearch, `paper/paper.pdf`.